

29 April 2026

Review Panel
International Collaboration in High-Performance Computing
Academy of Finland

Dear Colleagues:

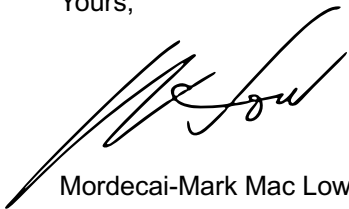
I am enthusiastic to collaborate with Maarit Korpi-Lagg and Frederick Gent on their project entitled "Extended multiscale code coupling for dust processing in supernova-driven turbulence" submitted in response to the 2026 call.

I am currently a tenured professor and curator in the Department of Astrophysics at the Gilder Graduate School of the American Museum of Natural History, as well as an adjunct professor in the Department of Astronomy of Columbia University, both in the United States. I am the PI on a US Department of Energy INCITE grant with Gent as col that is being used to run Pencil Code-Astaroth on the Frontier supercomputer, the largest publicly available machine in the country. I have published multiple papers with the PI and col on supernova-driven dynamos in the interstellar medium (Gent et al. 2020, 2021, 2023, 2024, 2025, 2026) as well as a review article in Living Reviews of Computational Astrophysics (Korpi-Lagg et al. 2024). I have also published on dust interactions with supernova remnants (Slavin et al. 2020) as well as extensively on the properties of the supernova-driven interstellar medium (e.g. Mac Low et al. 2001, 2005, de Avillez et al. 2002, Joung et al. 2006, Hill et al. 2012, 2018, Gatto et al. 2015). My research has been recognized with a Humboldt Research Prize and Fellowship in the American Astronomical Society.

With this letter I commit to fulfilling my responsibilities in the proposed collaboration of advising on interstellar medium and dust properties and processes, assisting in formulating science goals and interpreting results, and collaborating on publications. Our successful collaboration to date has been primarily implemented using weekly online video conversations as well as shared Git repositories for code and publications. I will suggest that the graduate student to be supported by this grant visit my institution to focus on interstellar medium properties and processes. I have found short visits of a month or so to be quite effective in past collaborations.

I am of course happy to provide any additional information that may be useful.

Yours,



Mordecai-Mark Mac Low